→ Class notes + Homework

Booklet DPP (H.C. Verma)

→ Revision before next class

→ H.W ⇒ Star mark good Q → later revision

→ Short notes/formula Book

→ Solution X

Lecture 1 → Basic Maths (Trignometry)
→ Vectors → Addition
→ subtraction

[Next Lecture
→ Resolution
→ Dot Product]

Trignometry

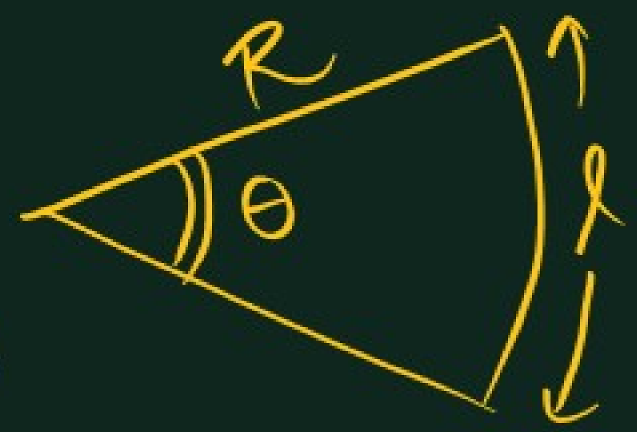
1) $30^\circ = 30 \times \frac{\pi}{180} = \frac{\pi}{6}$

→ Degree, radian

→ $\boxed{180^\circ = \pi \text{ radians.}}$

* $\boxed{\text{Degree value} \times \frac{\pi}{180} = \text{Radian value}}$

$$\left\{ \begin{array}{l} 30^\circ = \frac{\pi}{6}, \quad 45^\circ = \frac{\pi}{4}, \quad 60^\circ = \frac{\pi}{3} \\ 90^\circ = \frac{\pi}{2}, \quad 120^\circ = \frac{2\pi}{3}, \quad 180^\circ = \pi \end{array} \right.$$

2) 

$\boxed{\theta = \frac{l}{R} \text{ or } l = R\theta}$

θ = radians

(Complete circle, $\theta = 2\pi$
$\therefore l = R\theta = 2\pi R$)

3) Standard Ratios.

	0°	30°	45°	60°	90°
$\sin\theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan\theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined.

37° $\sin 37^\circ = 3/5$, $\cos 37^\circ = 4/5$, $\tan 37^\circ = 3/4$

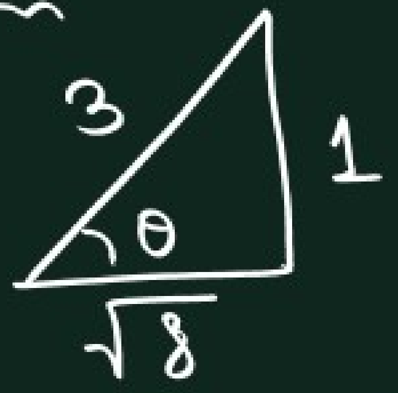
53° $\sin 53^\circ = 4/5$, $\cos 53^\circ = 3/5$, $\tan 53^\circ = 4/3$.

Q. $\sin\theta = \frac{1}{3}$

Find $\tan\theta = ??$

Ans: $\triangle$ method

$\tan\theta = \frac{1}{\sqrt{8}}$.



Q. $\tan\theta = 2/3$

Find $\sec\theta$.

$\cos\theta = \frac{3}{\sqrt{13}} \Rightarrow \sec\theta = \frac{\sqrt{13}}{3}$.

Identities

1) $\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta$

$\cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$

$\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta$

2) $\sin^2\theta + \cos^2\theta = 1$

3) $-1 \le \sin\theta \le 1$

$-1 \le \cos\theta \le 1$

4) $\boxed{\sin(2\theta) = 2\sin\theta\cos\theta}$

5) $\boxed{\cos 2\theta = 2\cos^2\theta - 1}$

$= 1 - 2\sin^2\theta$

$= \cos^2\theta - \sin^2\theta$

Similarly

$\left(\cos\theta = 2\cos^2(\theta/2) - 1\right)$

6) $\sin(A+B) = \sin A\cos B + \cos A\sin B$

$\sin(A-B) = \sin A\cos B - \cos A\sin B$

$\cos(A+B) = \cos A\cos B - \sin A\sin B$

$\cos(A-B) = \cos A\cos B + \sin A\sin B$

Q. Find 1) $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$

$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$

$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$

$\Rightarrow \left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)$

2) $\cos(120^\circ) = \cos(2 \times 60^\circ)$

$= 2\cos^2 60^\circ - 1$

$= 2 \times \frac{1}{4} - 1 = -\frac{1}{2}$

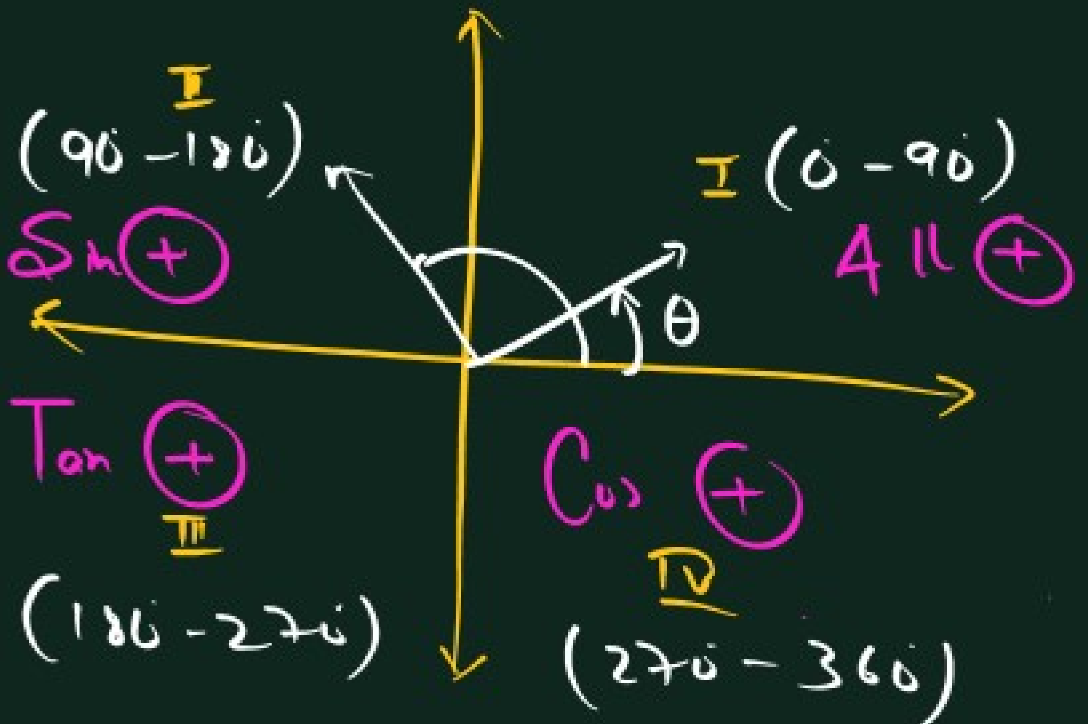
3) $\sin 120^\circ$

$= \sin(2 \times 60^\circ)$

$= 2\sin 60^\circ \cos 60^\circ$

$= \sqrt{3}/2$

→ Quadrants



A.S.T.C

$((180-\theta) \Rightarrow$ Quad. II $\Rightarrow \sin (+))$

$\sin(180° - \theta) = (+)\sin\theta$

$\cos(180° - \theta) = -\cos\theta$

$\tan(180° - \theta) = -\tan\theta$

Eg. $\cos(135°)$

$= \cos(180° - 45°)$

$= -\cos 45°$

$= -\frac{1}{\sqrt{2}}$

Eg. $\cos 150° = \cos(180° - 30°)$

$= -\cos 30°$

$= -\sqrt{3}/2$

$\cos 120° = -1/2$

$\cos 135° = -1/\sqrt{2}$

$\cos 150° = -\sqrt{3}/2$

$\cos 180° = -1$

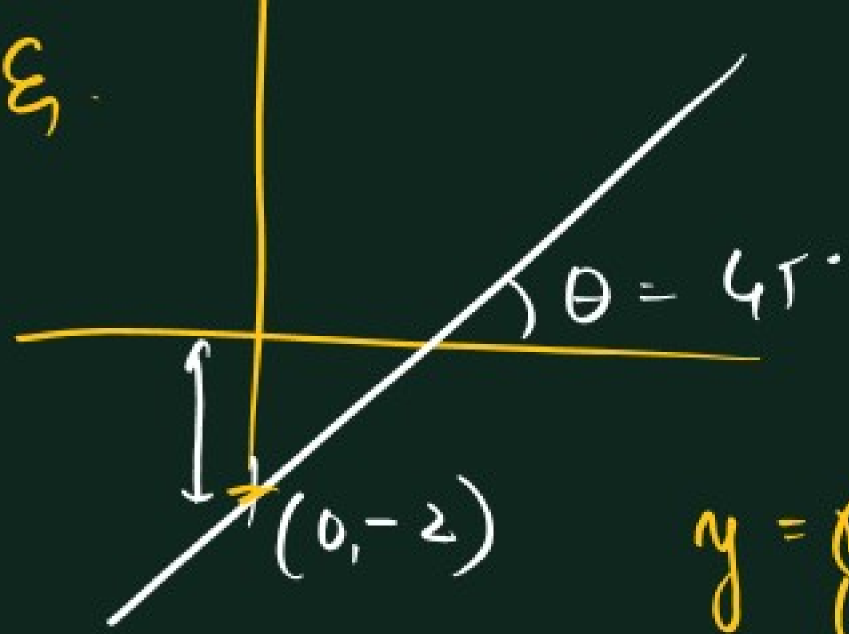
Equation of straight line.

$$\boxed{y = mx + c}$$

m = slope = $\tan\theta$

c = y-intercept.

Eg.



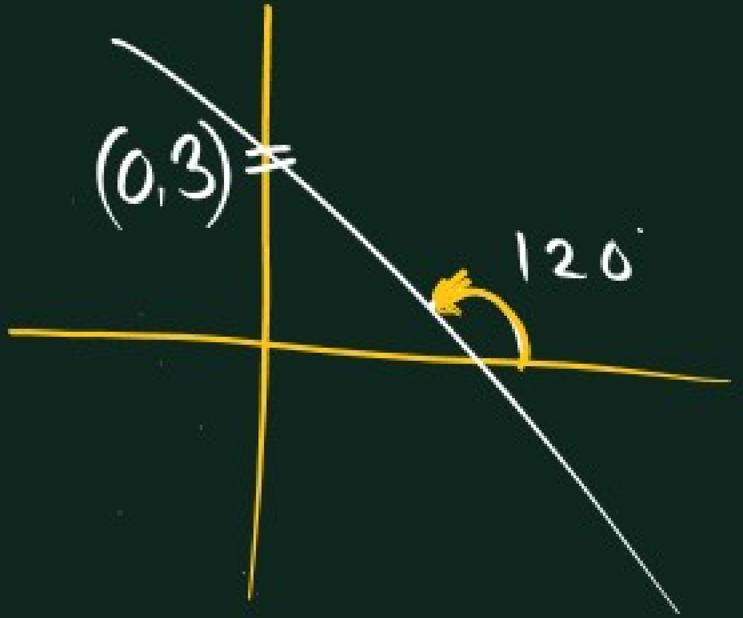
$y = mx + c$

$c = -2$

$m = \tan 45° = 1$

$\Rightarrow \boxed{y = x - 2}$

Eg.



$y = mx + c$

$m = \tan 120°$

$= \tan(180° - 60°)$

$= -\tan 60°$

$= -\sqrt{3}$

$\Rightarrow y = -\sqrt{3}x + 3$

Vectors.

Physical Quantities

Scalar

Eg. speed, distance, current, mass, pressure etc.

4 kg

- Magnitude only
- Follow normal algebraic addition

Vector

- velocity, displacement, Force, momentum, acceleration

3m/s in East direction

- Magnitude + Direction
- Vector rules of Addition (Triangle law)

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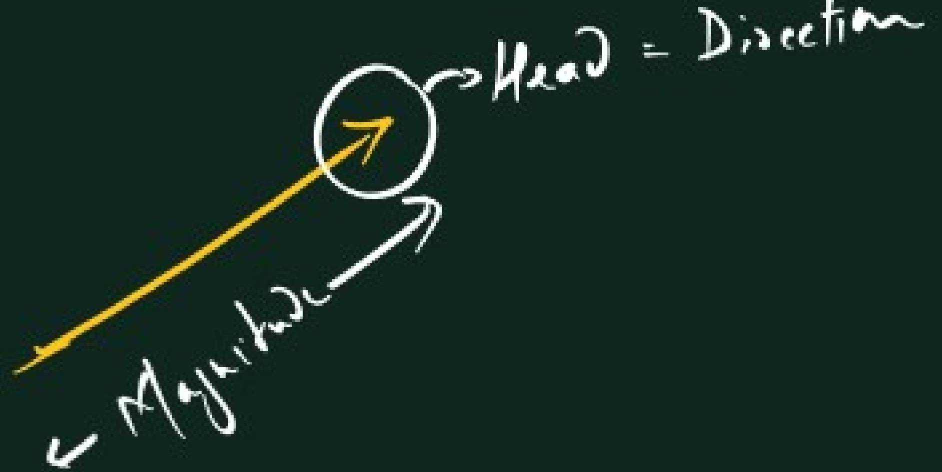
3 kg. Aloo
4 kg. Tamatar.

7 kg.

3 N
4 N

~~7 N~~

1)



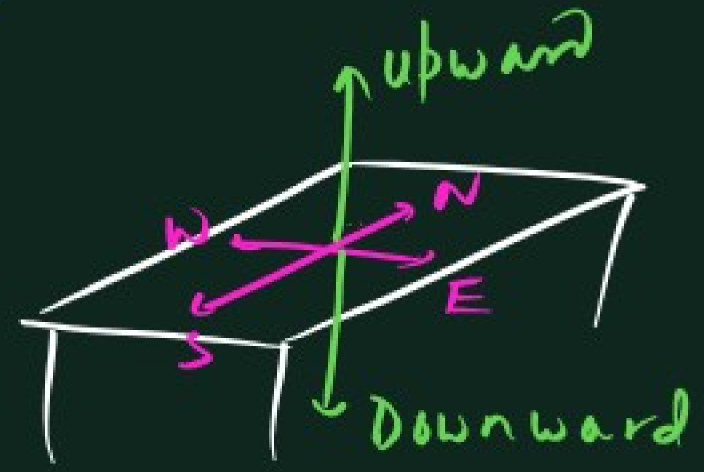
2) Symbol → $\vec{A}$, $\vec{a}$, $\vec{AB}$

3) Magnitude of vector. $|\vec{F}| = 5N$ or $F = 5N$

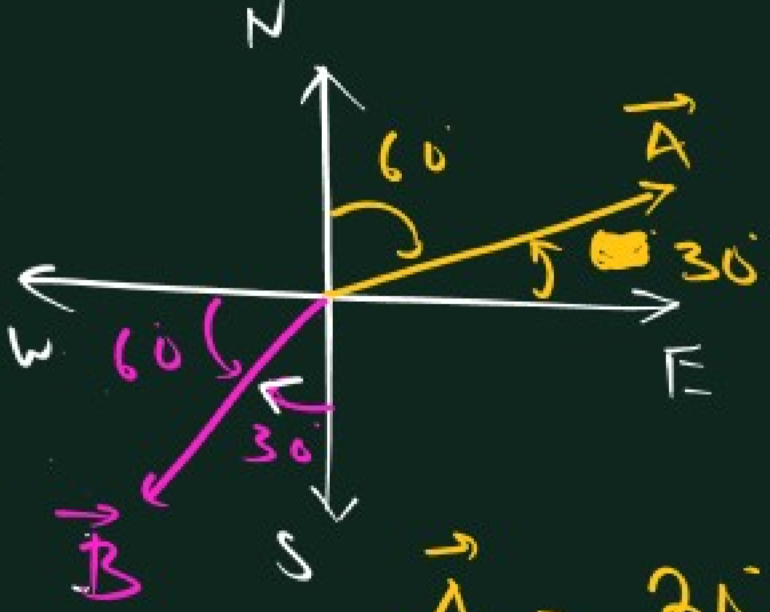
Modulus/ letter without arrow

Note: Direction

1)



2)

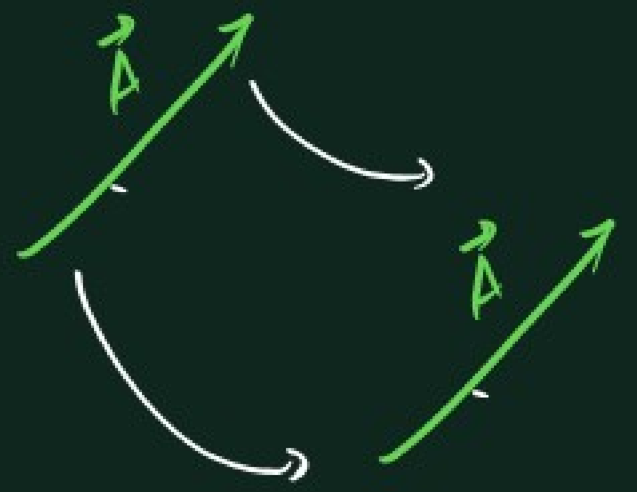


$\vec{A}$ = 30° North of East

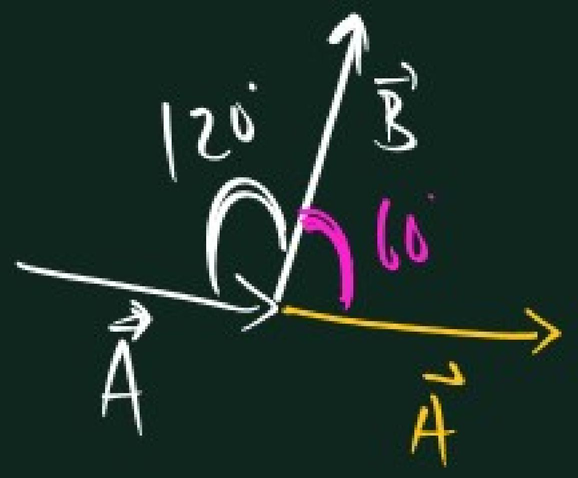
or $\vec{A}$ = 60° East of North

$\vec{B}$ = 30° W of S or 60° S of W

1)

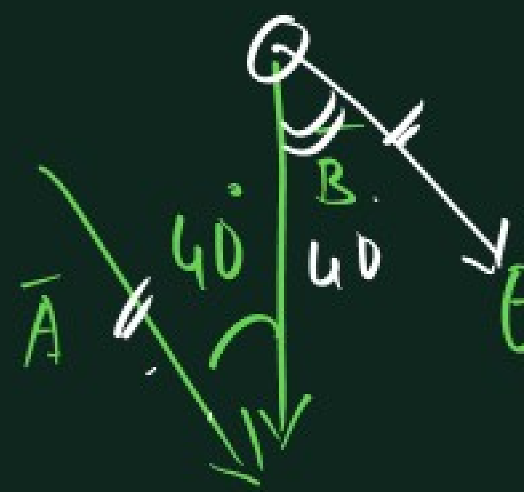


2) Angle between 2 vectors



$\underline{\underline{\theta = 60°}}$

Q



$\theta = ??$

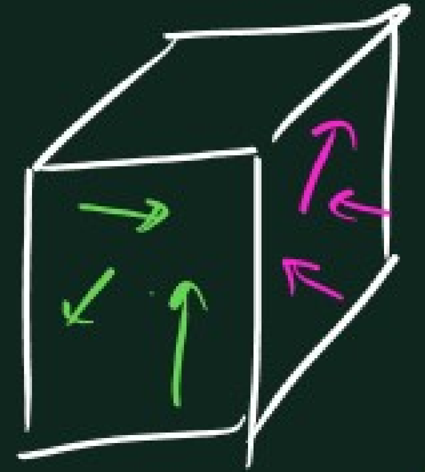
3) Types of vectors.

1)

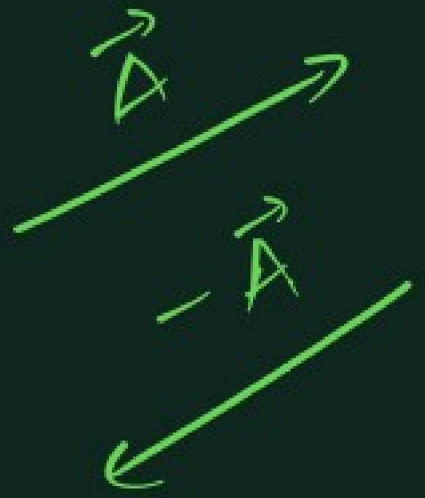
2)

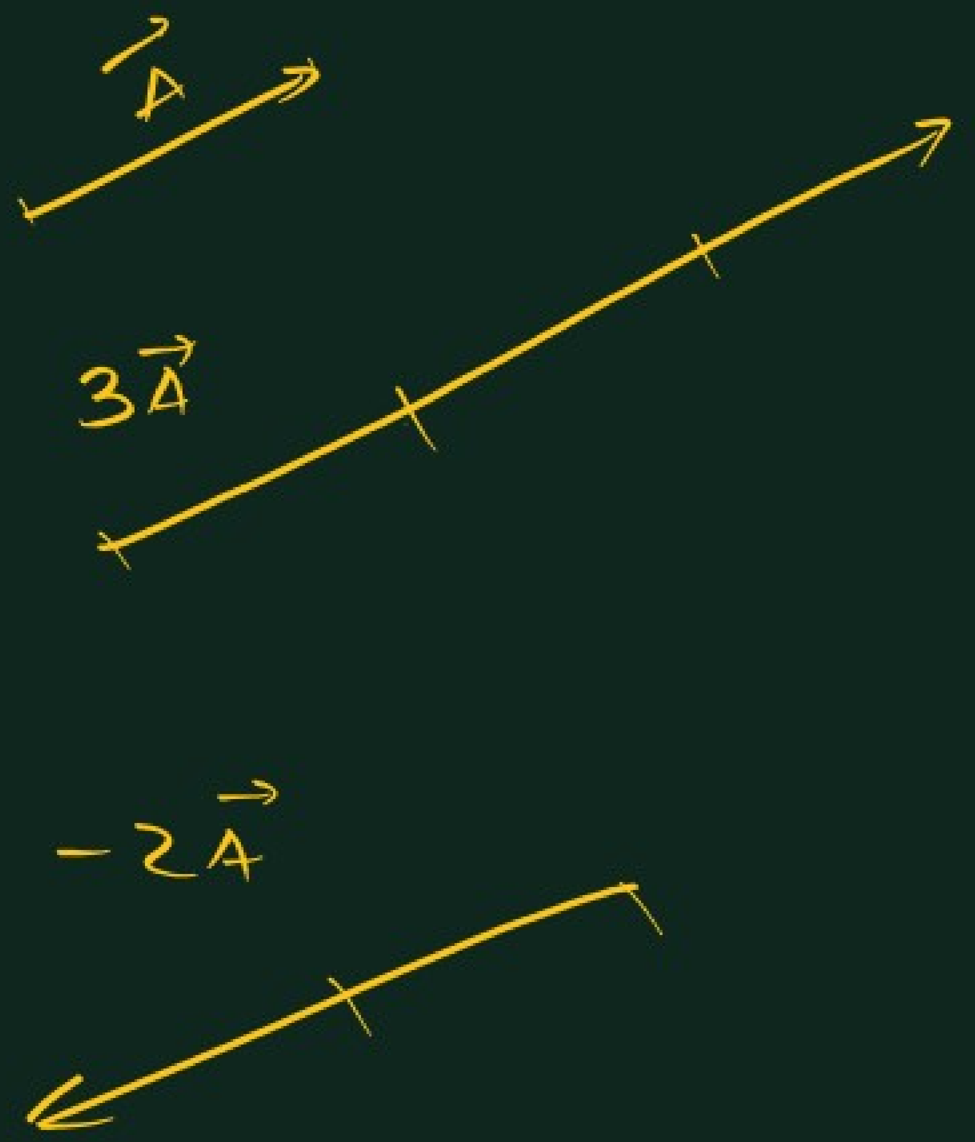
3)

4)



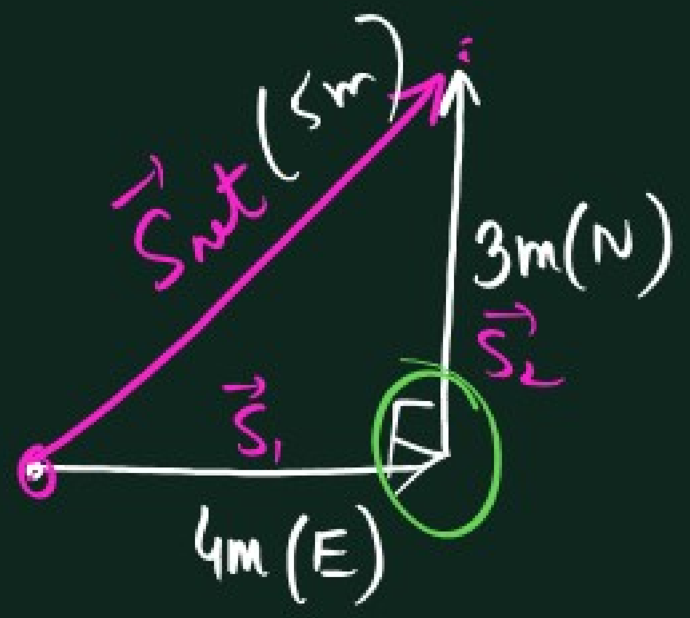
5)





$\vec{A}$
$3\vec{A}$
$-2\vec{A}$

Triangle law.

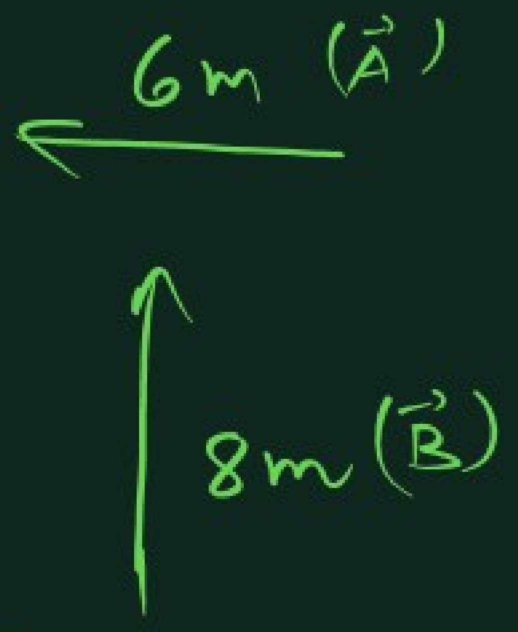


$\bar{S}_1 = 4m(E)$

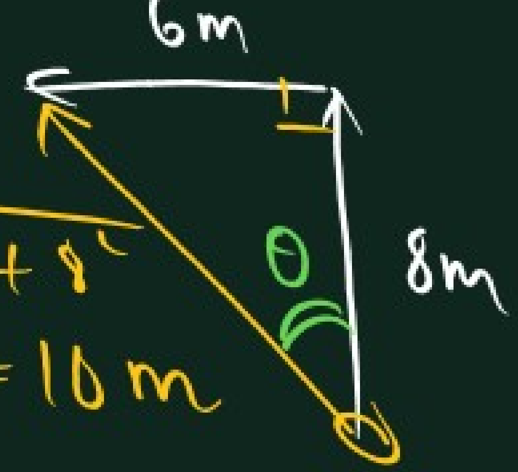
$\bar{S}_2 = 3m(N)$

$\bar{S}_{net} = \bar{S}_1 + \bar{S}_2 = 5m(37° \text{ N of E})$

Q1.



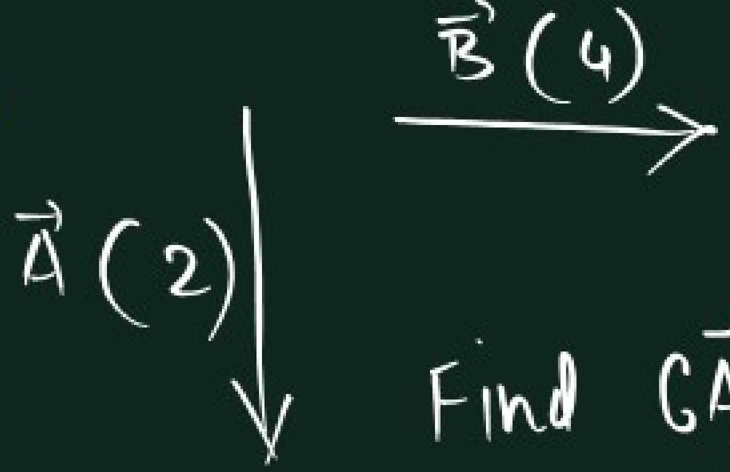
$\vec{A} + \vec{B} = ??$



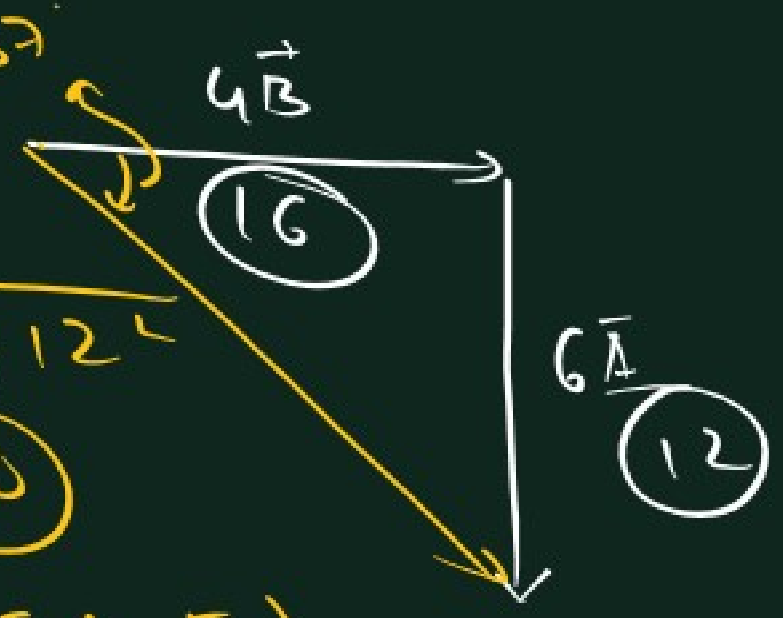
$\sqrt{6^2+8^2} = 10m$

$\tan\theta = 3/4 \Rightarrow \theta = 37°$ W of N

Q2)



Find $6\vec{A} + 4\vec{B}$.



$\sqrt{16^2+12^2} = 20$

(37° S of E)

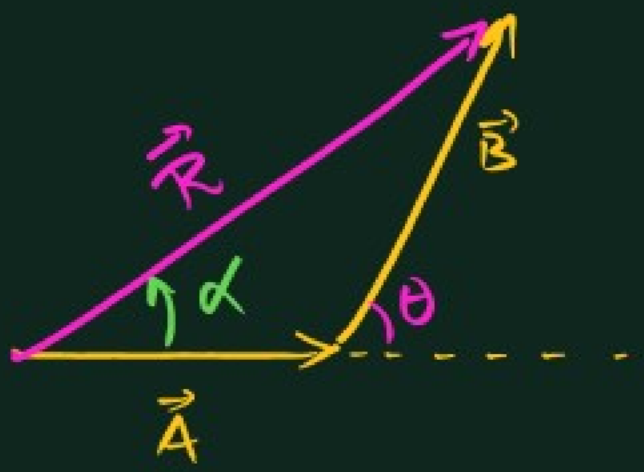
Note: **Pythagorean Triplets**

1) $(3, 4, 5)$, $(6, 8, 10)$, $(9, 12, 15)$, $(12, 16, 20)$ ...

2) $(5, 12, 13)$

3) $(8, 15, 17)$

Formula



$(\vec{R} = \vec{A} + \vec{B})$

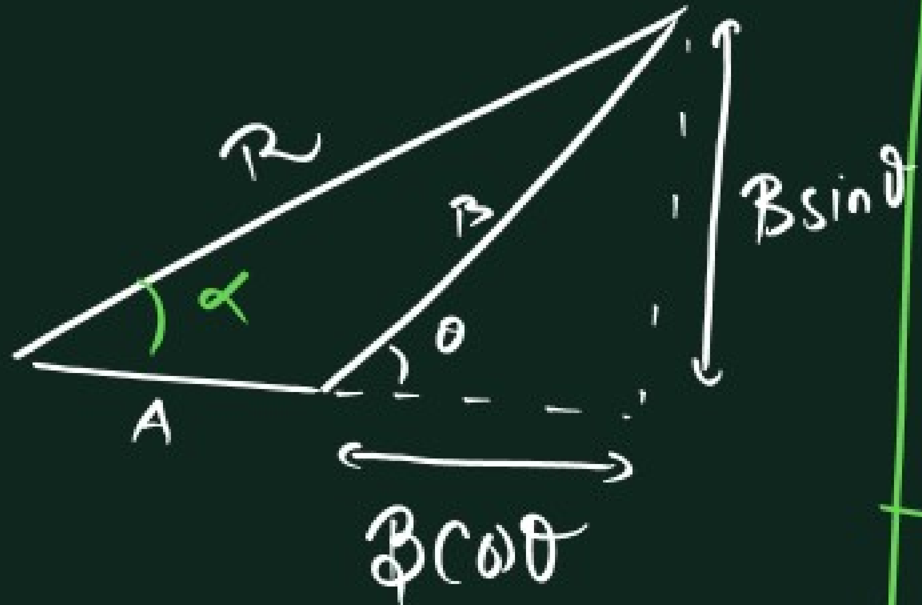
To find : Magnitude & direction of $\vec{R}$ in terms of (A, B, θ)

$\sin\theta = P/H$

$\Rightarrow P = H\sin\theta$

$\cos\theta = B/H$

$\Rightarrow B = H\cos\theta$



By Pythagoras theorem

$$R = \sqrt{(A + B\cos\theta)^2 + (B\sin\theta)^2}$$

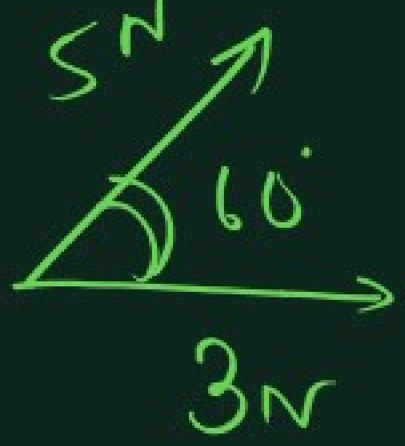
$$= \sqrt{A^2 + B^2\cos^2\theta + 2AB\cos\theta + B^2\sin^2\theta}$$

$$\Rightarrow \boxed{R = \sqrt{A^2 + B^2 + 2AB\cos\theta}}$$

Angle made by $\vec{R}$ with $\vec{A}$

$$\boxed{\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}}$$

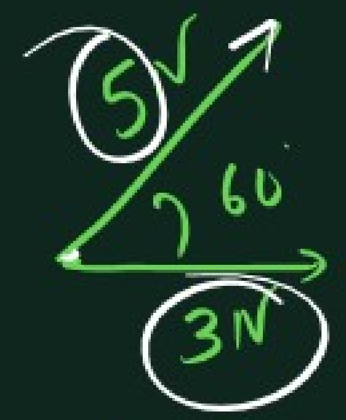
Q1.



Find 1) net force

2) angle of net force with 3N

a.



1) Net F

2) Angle with 3N

$R=\sqrt{A^2+B^2+2AB\cos\theta}$

$A=3$

$B=5$

$\theta=60^\circ$

1) $R=\sqrt{3^2+5^2+2\times3\times5\times\frac{1}{2}}$

$=\sqrt{49}=7N$

2) Angle of $\vec{R}$ with 3N ($\vec{A}$)

$$\tan\alpha=\frac{B\sin\theta}{A+B\cos\theta}$$

$$\tan\alpha=\frac{5\times\sqrt{3}/2}{3+5\times\frac{1}{2}}$$

$\Rightarrow \tan\alpha=\frac{5\sqrt{3}}{11}$ or $\boxed{\alpha=\tan^{-1}\left(\frac{5\sqrt{3}}{11}\right)}$

A) 60° B) $\cos\theta=1/4$

C) $\cos\theta=3/8$ D) None

$A=2,\ B=3,\ R=4$

$4=\sqrt{2^2+3^2+2\times3\times2\times\cos\theta}$

$\Rightarrow 16=13+12\cos\theta$

$\Rightarrow \boxed{\cos\theta=1/4}$

Q. A, A $\Rightarrow$ A

A: $R^2 = A^2 + B^2 + 2AB\cos\theta$

$R = A = B$

$A^2 = \underbrace{A^2 + A^2} + 2A^2\cos\theta$

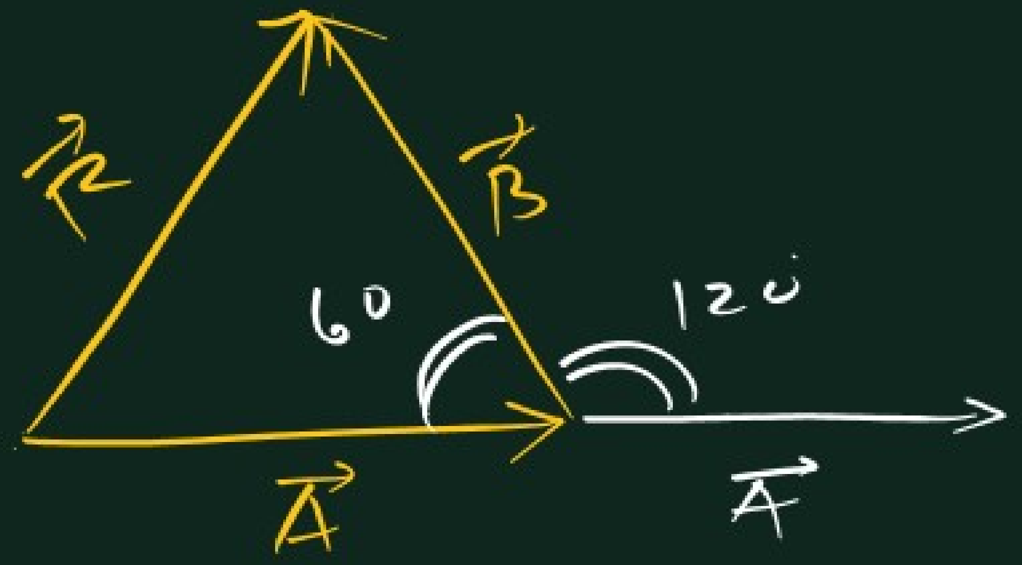
$\Rightarrow A^2 - 2A^2 = 2A^2\cos\theta$

$\Rightarrow -\frac{\cancel{A^2}}{2\cancel{A^2}} = \cos\theta$

$\Rightarrow \cos\theta = -\frac{1}{2} \Rightarrow \boxed{\theta = 120^\circ}$

2^{nd} way

Equilateral Δ



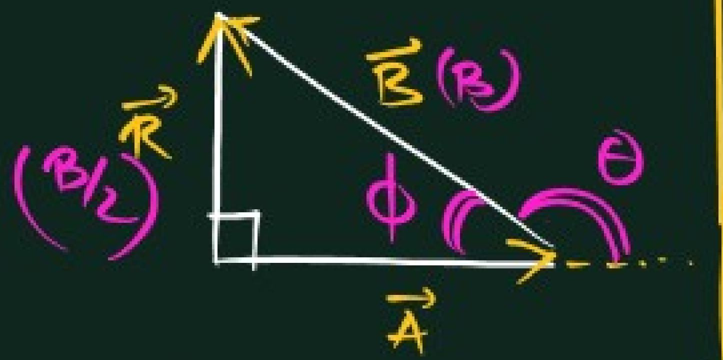
($\vec{A}, \vec{B}, \vec{R} \Rightarrow \Delta$ formed)

$\theta = 120^\circ$

Q. $\vec{R} \perp \vec{A}$

$R = \frac{1}{2} B$

$\theta(\vec{A}, \vec{B}) = ??$



$\sin\phi = \frac{B/2}{B} = 1/2$

$\Rightarrow \phi = 30^\circ$

$\Rightarrow \boxed{\theta = 150^\circ}$

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$\cos\theta = 1$ (max) for $\theta = 0^\circ$

$R_{max} = \sqrt{A^2 + B^2 + 2AB}$

$= \sqrt{(A+B)^2}$

$\Rightarrow R_{max} = A + B \quad (\theta = 0^\circ)$

$\cos\theta = -1$ (min)

for $\theta = 180^\circ$

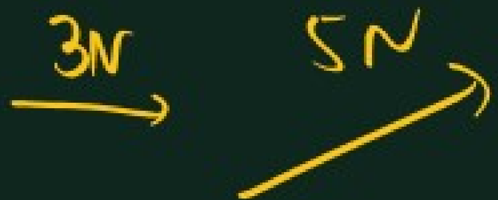
$R_{min} = \sqrt{A^2 + B^2 + 2AB(-1)}$

$= \sqrt{(A-B)^2}$

$\rightarrow R_{min} = |A - B|$

$(\theta = 180^\circ)$

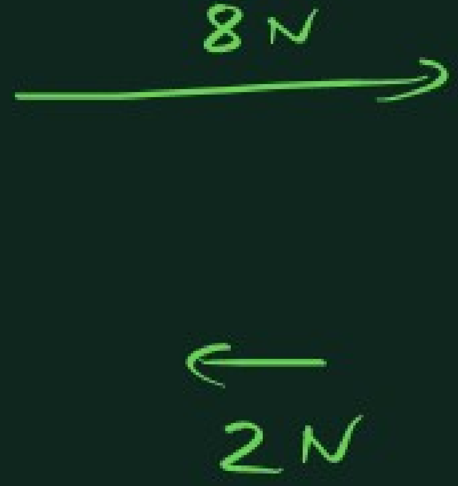
Eg



Ans $R_{max} = 3 + 5 = 8N$ $\xrightarrow{3}\xrightarrow{5}$

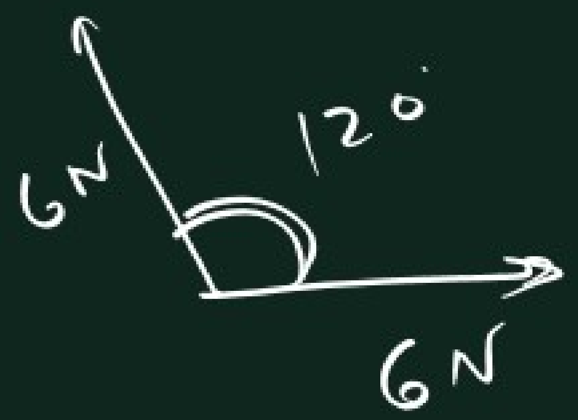
$R_{min} = 5 - 3 = 2N$ $\xrightarrow{5}\xleftarrow{3}$

$|A - B| \le R \le A + B$



$R = \sqrt{A^2 + B^2}$

$\boxed{R = 2A\cos(\theta/2)}$



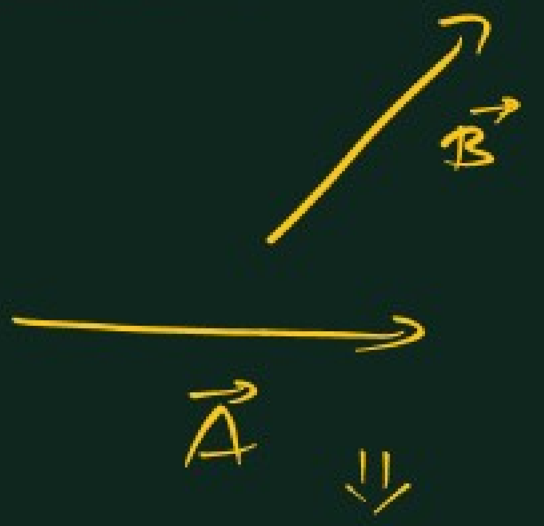
$\Downarrow$

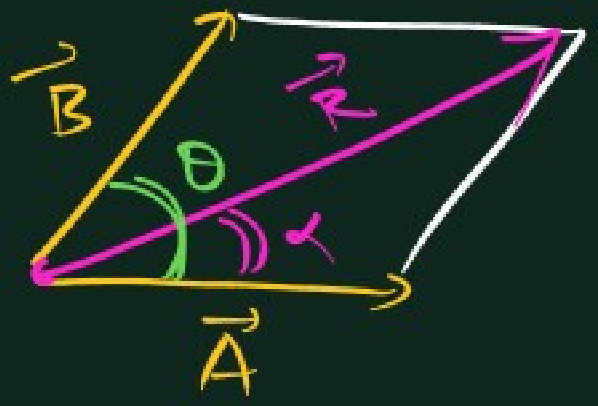
$$R = 2A\cos(\theta/2)$$
$$= 2 \times 6 \times \cos 60°$$
$$= 6$$

H.W

Show $R = 2A\cos(\theta/2)$ if (A, A) are added

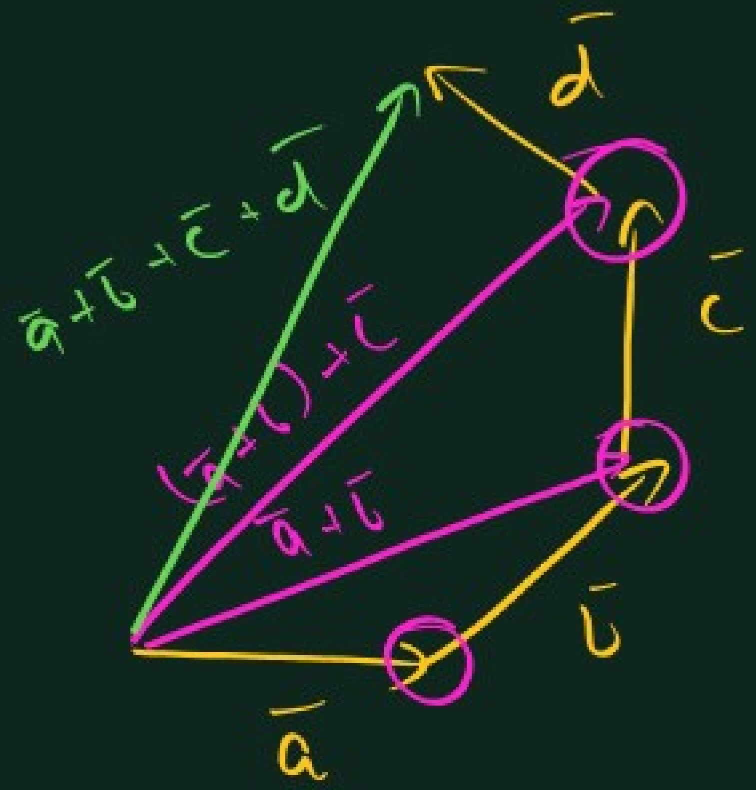
Parallelogram law

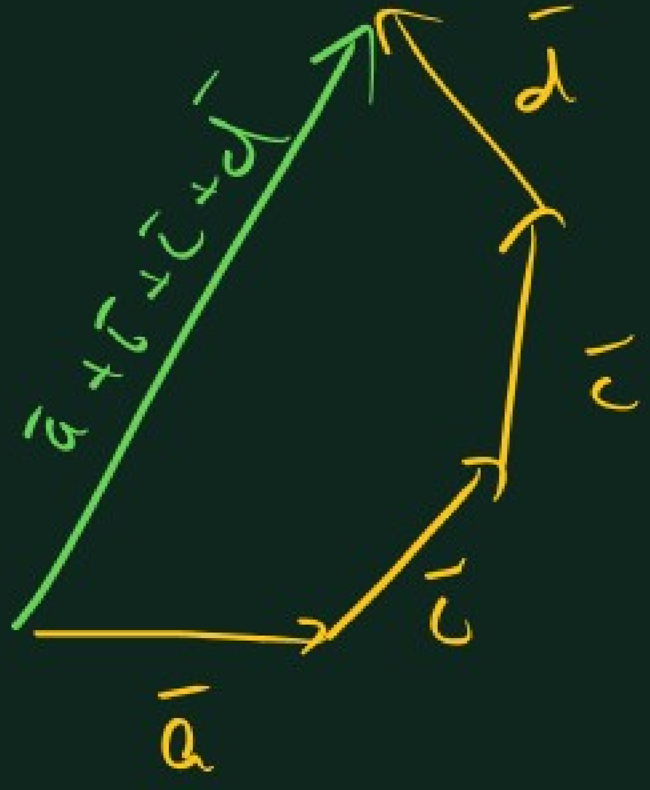




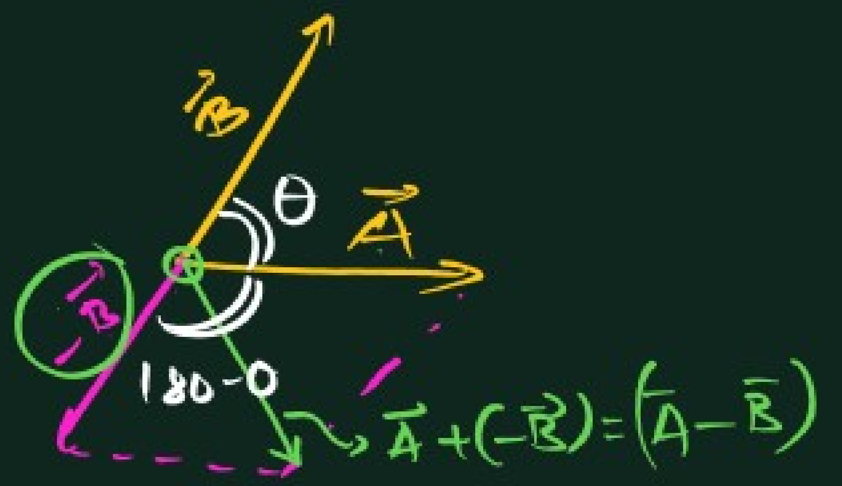
$$R=\sqrt{A^2+B^2+2AB\cos\theta}$$

$$\tan\alpha = \frac{B\sin\theta}{A+B\cos\theta}$$





Subtraction of 2 vectors

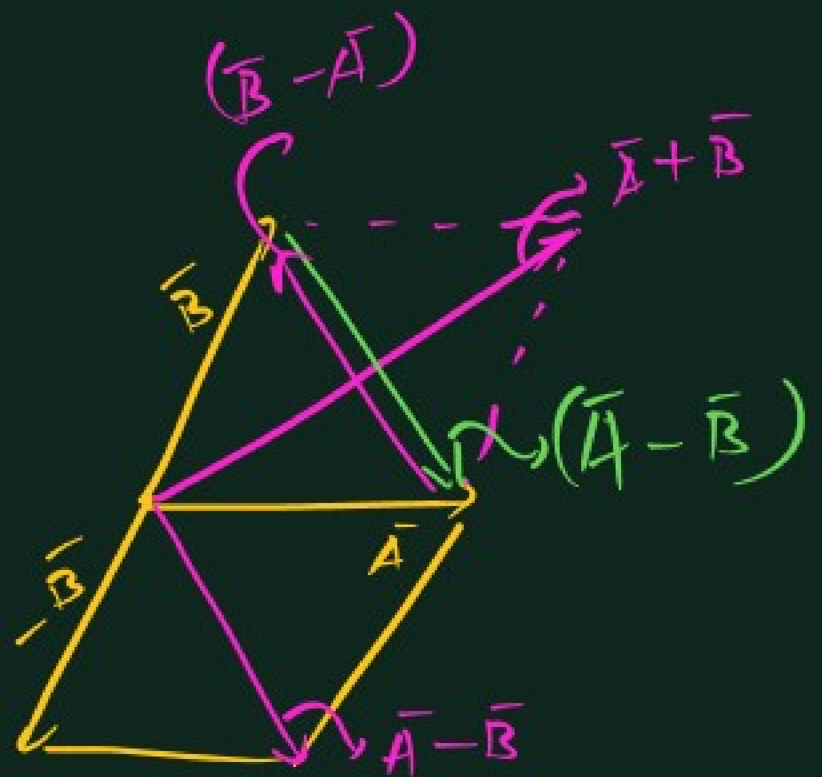


$$\vec{A}-\vec{B}=\vec{A}+(-\vec{B})$$

$$|\vec{A}+(-\vec{B})|=\sqrt{A^2+B^2+2AB\underbrace{\cos(180-\theta)}_{-\cos\theta}}$$

$$\boxed{|\vec{A}-\vec{B}|=\sqrt{A^2+B^2-2AB\cos\theta}}$$

θ = angle between $\vec{A}$ & $\vec{B}$
(not $\vec{A}$ & $-\vec{B}$)



2 diagonals of ||gram → 1 = sum
→ 1 = difference.

→ Head of $\bar{A}-\bar{B}$ with $\bar{A}$
→ Head of $\bar{B}-\bar{A}$ with $\bar{B}$

Q1) If 2 forces of 4N & 6N are applied, which of following value(s) of resultant is possible

~~A) 1N~~   B) 5N
C) 10N   ~~D) 15N~~

$$\underset{=2}{6-4}\le|\vec{A}+\vec{B}|\le\underset{=10}{4+6}$$

Q2. $|\vec{A}| = 5N$

$|\vec{B}| = 10N$

Angle b/w them = 37°

Find $|\vec{A}-\vec{B}|$??

Ans: $|\vec{A}-\vec{B}| = \sqrt{A^2+B^2-2AB\cos\theta}$

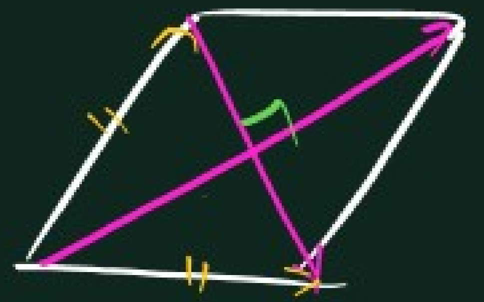
$= \sqrt{5^2+10^2-2\times5\times10\times\frac{4}{5}}$

$= \sqrt{45}$ or $3\sqrt{5}$

Q3. If $(\vec{A}+\vec{B}) \perp (\vec{A}-\vec{B})$

(perpendicular)

find relation b/w $|\vec{A}|$ & $|\vec{B}|$



$\Downarrow$

Rhombus = Diagonals $\perp^r$

$\Rightarrow |\vec{A}| = |\vec{B}|$

Q4) If $|\vec{A}+\vec{B}| = |\vec{A}-\vec{B}|$ → Diagonals equal

$\Rightarrow$ find θ b/w $\vec{A}$ & $\vec{B}$

$\Rightarrow$ Rectangle

$\Rightarrow \vec{A} \perp^r \vec{B}$

Parallelogram

Rhombus — Diagonals $\perp^r$

Rectangle — Diagonals equal

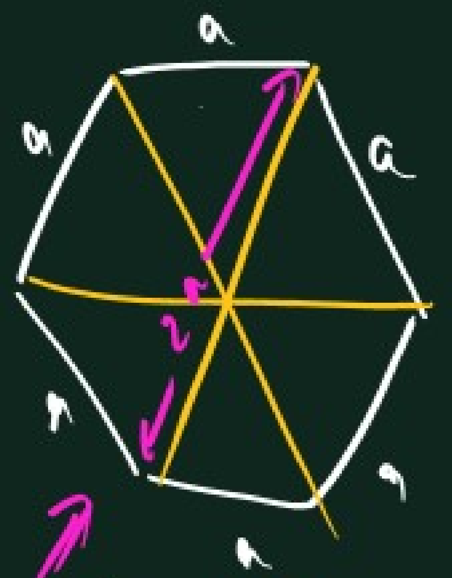
Square

A) 6F   B) 0

C) 2F   D) F

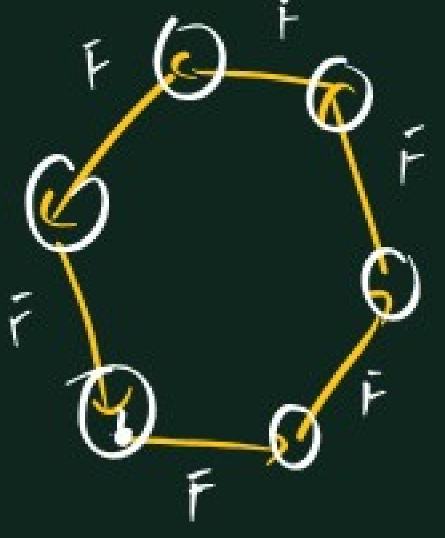
Regular hexagon.

All sides equal



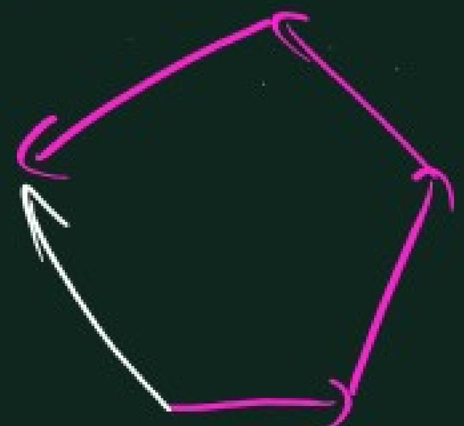
6 equilateral Δs.

i)

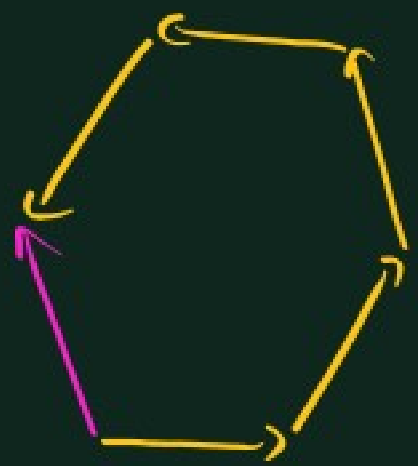


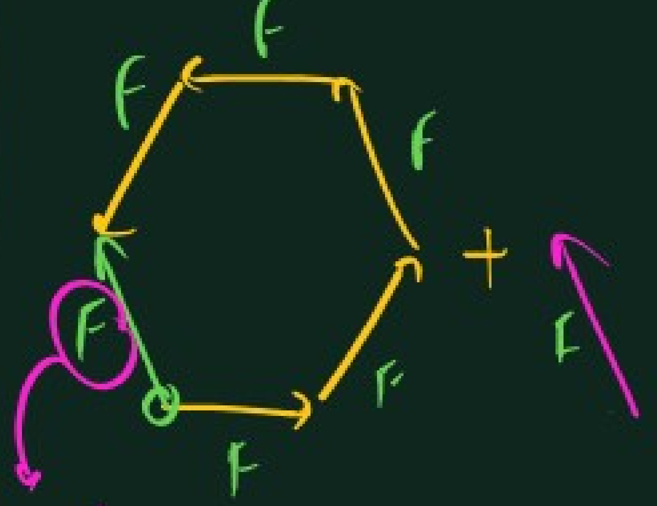
$\Downarrow$

Resultant = 0



ii)





By polygon law sum of the 5 forces

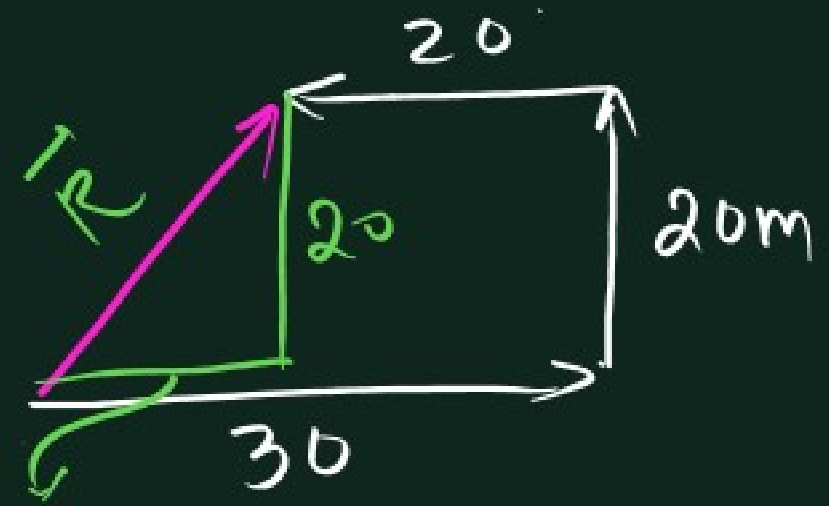
$\Downarrow$

Q. Man walking.

30m E, turns left (90°) & 20m, turns left & 20m

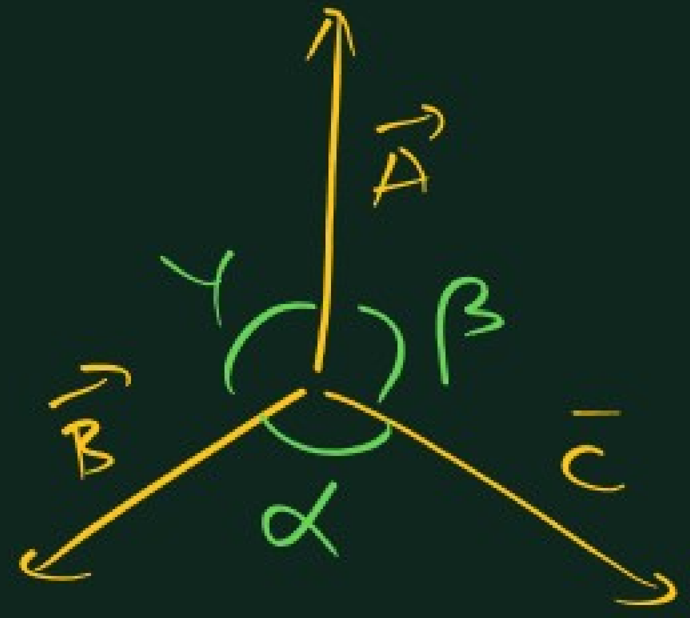
Net displacement

Ans:



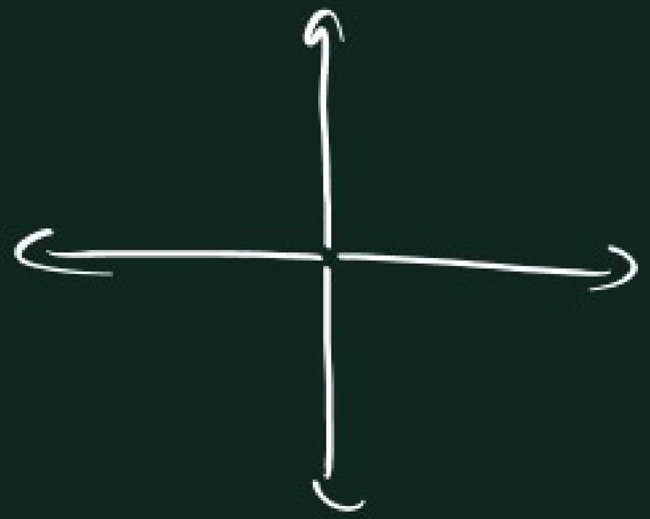
$|\vec{R}| = \sqrt{20^2 + 10^2}$

$= 10\sqrt{5}$



$$\left(\frac{360}{n}\right)$$

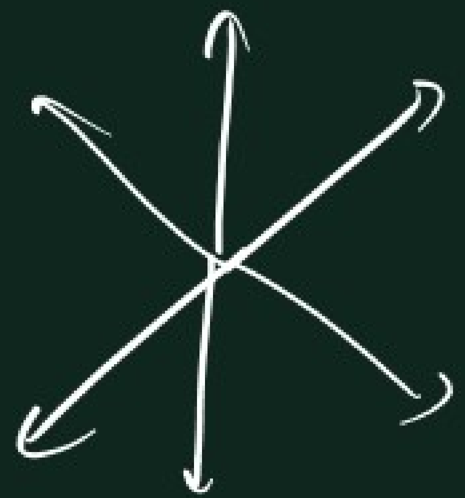
Eg. 4 vectors $\Rightarrow \frac{360}{4} = 90$



$$\boxed{\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}}$$

Lamis theorem

6 $\Rightarrow \theta = \frac{360}{6} = 60$



A) 10, 10, 10 → Δ possible ⇒ R can be 0

B) 10, 10, 20 →

C) 10, 20, 23 → $10+20 > 23$

D) 10, 20, 40 ✓ → $\boxed{10+20 \not> 40}$

Δ not possible ⇒ $R \neq 0$

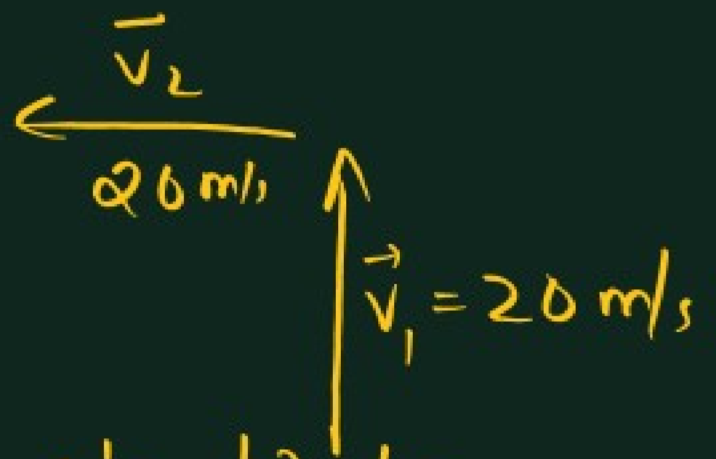
3 sides of Δ

Sum of 2 sides $\geq 3^{rd}$ side

H.W

Q1) If sum of 2 unit vectors is a unit vector, find magnitude of their difference. $(\sqrt{3})$

Q2)



Magnitude of change in velocity?

$[20\sqrt{2}$ S-W$]$

- Intext 4
- Ex 1 → 38 - 69
- Ex 2 → 48 - 72